Dr. Malik Abdul Manan

Passport: AD0704843 | **Work permit:** Chinese | **Date of birth:** 27/10/1992 | **Place of birth:** Dera Ghazi Khan, Punjab, Pakistan | **Nationality:** Pakistani | **Phone:** (+86) 15010727593 (Mobile) | **Email:** manansandila@yahoo.com | **Website:** https://www.researchgate.net/profile/Malik-Manan-2 | **Website:** https://scholar.google.com/citations?user=HFZqAYsAAAAJ&hl=en | **LinkedIn:** https://www.linkedin.com/in/malik-abdul-manan-sandila/ | **Whatsapp Messenger:** +923366940060 | **Address:** Beijing University of Technology , 100124, Beijing, China (Work)

ABOUT MYSELF

As a PhD researcher with 20+ publications in artificial intelligence-based medical image analysis, specializing in detecting polyps, tumors, and exudates, I am now actively expanding my deep learning expertise into industrial, agricultural, and bioinformatics applications. My proven capabilities in developing high-precision diagnostic algorithms—including CNN architectures, attention mechanisms, and parallel feature extraction—provide a rigorous foundation for adapting these methods to industrial manufacturing (defect detection/classification), agricultural technology (crop disease identification/monitoring), and genomic analysis (biomarker identification). I am particularly interested in leveraging my medical imaging experience to develop cross-domain AI solutions where diagnostic-grade accuracy can solve critical challenges in medicine, industrial automation, and sustainable agriculture.
ORCID ID: 0009-0007-8644-4655
Scopus ID: 57423411100
Web of Science Researcher ID: JGM-3316-2023

EDUCATION AND TRAINING

02/09/2020 – 09/06/2025 Beijing, China
PHD (ELECTRONIC SCIENCE AND TECHNOLOGY) Beijing University of Technology

Website https://english.bjut.edu.cn/ | **Field of study** Information and Communication Technologies (ICTs) |

Level in EQF EQF level 6 |

Thesis Deep Learning Scheme-Based Study on Multi-Modal Medical Image Analysis for Improved Diagnosis

02/02/2017 – 31/01/2019 Islamabad, Pakistan
MASTER OF ELECTRICAL ENGINEERING (COMPUTER) COMSATS University Islamabad

Website https://www.comsats.edu.pk/ | **Level in EQF** EQF level 6

02/02/2012 – 31/01/2016 Islamabad, Pakistan
BACHELOR OF ELECTRICAL (COMPUTER) ENGINEERING COMSATS University Islamabad

Website https://www.comsats.edu.pk/ | **Level in EQF** EQF level 6

MATHWORKS COMPUTER VISION ENGINEER MathWorks

Website https://coursera.org/share/450ffcfd836caae243fd553525846225 | **Level in EQF** EQF level 6

IBM AI ENGINEERING IBM

Website https://coursera.org/share/1c0799cacfebd470e4a6f6ca1a575707 | **Level in EQF** EQF level 6

PUBLICATIONS

2024

Advancement in Semantic Segmentation Techniques: A Comprehensive Review for Semantic Segmentation of Colorectal Polyps Using Deep Learning (Accepted)

(SCI Q1, Impact factor=13.9)

Malik Abdul Manan, Feng Jinchao, Shahzad Ahmed, Artificial Intelligence Review

2025

LWAH-Net: Light weight Attention-Driven Hybrid Network for Polyp Segmentation in Endoscopic Images

(SCI Q1, Impact factor=6.8)

Malik Abdul Manan, Feng Jinchao, Shahzad Ahmed, Abdul Raheem

2024

Multi-scale and multi-path cascaded convolutional network for semantic segmentation of colorectal polyps

(SCI Q1, Impact factor=6.8)

Malik Abdul Manan, Jinchao Feng , Muhammad Yaqub, Alexandria Engineering Journal

2025

Enhancing Colorectal Polyp Segmentation with TCFMA-Net: A Transformer-Based Cross Feature and Multi-Attention Network

(SCI Q1, Impact factor=6.2)

Malik Abdul Manan, Feng Jinchao, Shahzad Ahmed, Abdul Raheem, Artificial Intelligence in Medicine

2025

MSCR-Net: Multipath Separable Convolution-based Residual Network for Segmentation of Colorectal Polyp (Revision)

(SCI Q1, Impact factor=4.9)

Malik Abdul Manan, Feng Jinchao, Shahzad Ahmed, Abdul Raheem, Biomedical Signal Processing and Control

2025

Multiscale Feature Fusion Booster Network for Segmentation of Colorectal Polyp

(SCI Q2, Impact factor=3.0)

Malik Abdul Manan, Jinchao Feng, Shahzad Ahmed, Abdul Raheem, International Journal of Imaging Systems and Technology

2023

Semantic segmentation of retinal exudates using a residual encoder–decoder architecture in diabetic retinopathy

(SCI Q2, Impact factor=2.89)

Malik Abdul Manan, Feng Jinchao, Tariq M. Khan, Microscopy Research and Technique

2024

DPE-Net: Dual-Parallel Encoder Based Network for Semantic Segmentation of Polyps

(EI Conference)

Malik Abdul Manan, Feng Jinchao, Shahzad Ahmed, Abdul Raheem, 9th International Conference on Signal and Image Processing (ICSIP)

2025

FedGraphMRI-net: A federated graph neural network framework for robust MRI reconstruction across non-IID data

(SCI Q1, Impact factor=4.9)

Shahzad Ahmed, Feng Jinchao, Malik Abdul Manan, Muhammad Yaqub, Biomedical Signal Processing and Control

2023

DeepLabV3, IBCO-based ALCResNet: A fully automated classification, and grading system for brain tumor

(SCI Q1, Impact factor=6.8)

Muhammad Yaqub, Feng Jinchao, Shahzad Ahmed, --------, Malik Abdul Manan

2024

ADLER-MRI: Adaptive Deep Learning for Enhanced MRI Reconstruction with Noise-Resilient Models

(EI Conference)

Shahzad Ahmed; Feng Jinchao; Malik Abdul Manan, IEEE International Symposium on Biomedical Imaging

2023

Optimizing Fully Convolutional Encoder-Decoder Network for Segmentation of Diabetic Eye Disease

(SCI Q2, Impact factor=2.1)

Abdul Qadir Khan, Guangmin Sun, Yu Li, Anas Bilal, Malik Abdul Manan, Computers, Materials&Continua

2025

PINN-DADif: Physics-Informed Deep Adaptive Diffusion Network for Robust and Efficient MRI Reconstruction

(SCI Q2, Impact factor=2.9)

Shahzad Ahmed, Feng Jinchao, Atif Mehmood, --------, Malik Abdul Manan, Digital Signal Processing

2025

GraFMRI: A graph-based fusion framework for robust multi-modal MRI reconstruction

(SCI Q2, Impact factor=2.1)

Shahzad Ahmed, Feng Jinchao, Javed Ferzund, ---------, Malik Abdul Manan, Magnetic Resonance Imaging

2024

IPC-CNN: A Robust Solution for Precise Brain Tumor Segmentation Using Improved Privacy-Preserving Collaborative Convolutional Neural Network

(SCI Q3, Impact factor=1.2)

Abdul Raheem, Zhen Yang, Haiyang Yu, Muhammad Yaqub, --------, Malik Abdul Manan, KSII Transactions on Internet and Information Systems (TIIS)

2025

FAME: A Federated Adversarial Learning Framework for Privacy-Preserving MRI Reconstruction

(SCI Q3, Impact factor=1.1)

Shahzad Ahmed, Jinchao Feng, Javed Ferzund, ----------, Malik Abdul Manan, Applied Magnetic Resonance

WORK EXPERIENCE

BDIC BEIJING-DUBLIN INTERNATIONAL COLLEGE – BEIJING, CHINA
Business or Sector Education

TEACHING ASSISTANT IN UNIVERSITY – 01/09/2024 – 30/05/2025

Facilitated discussions and addressed student inquiries to enhance understanding of course content.

RIPHAH INTERNATIONAL COLLEGE – D.G.KHAN, PAKISTAN

DEPARTMENT COORDINATOR (ADMISSIONS MANAGER) – 02/07/2020 – 31/12/2022

Worked as an Admissions manager and was in charge of the students' applications and admissions by assessing students' qualifications and subsequently approving or denying their applications based on the regulations set by the board of directors and the college administration. Worked as a department coordinator of Computer science and Information technology.

·Teaching Subjects: Bioinformatics, Digital Logic Design, Artificial Intelligence, Computer application, Electronics
·Conducting Labs: Matlab, Digital Logic Design (Proteus, Modelsim), LTSpice Orcad

UNIVERSITY OF EDUCATION – D.G.KHAN, PAKISTAN

LECTURER (VISITING) – 02/02/2019 – 06/03/2021

·Teaching Subjects: Digital Logic Design, Bioinformatics, Electronics
·Conducting Labs: Digital Logic Design (Proteus, Modelsim)

MIR CHAKAR KHAN RIND UNIVERSITY – D.G.KHAN, CHINA

LECTURER (VISITING) – 01/09/2019 – 31/08/2020

·Teaching Subjects: Digital Logic Design,Discrete Structure,Micro-Processor,Computer application
·Conducting Labs: Digital Logic Design (Proteus ,Modelsim)

COMSAT UNIVERSITY ISLAMABAD – ISLAMABAD, PAKISTAN

NATIONAL INTERNSHIP PROGRAM – 01/07/2017 – 30/06/2018

·Doing research work in the field of computer vision and deep learning. Took some visiting lectures and conducted labs.

THE SPECTRUM SCHOOL & COLLEGE – ISLAMABAD, PAKISTAN

LECTURER – 01/03/2016 – 30/05/2017

·Developed and implemented daily and unit lesson plans for SSC and HSSC, administered tests and evaluations, and analyzed student performance.
·Teaching subjects are Physics, Mathematics, and Computer science.

SKILLS

MATLAB | Python | Deep Learning | Software and Programming | Microsoft Visio | Research paper writing | Proteus | Analysis | Communication | Microsoft Office | Adobe Photoshop | Problem Solving | Draw IO | LTSpice | Image Processing | Computer Vision

HONOURS AND AWARDS

24/05/2025
Science and Technology Innovation Award Beijing University of Technology (Three Awards) – Beijing University of Technology

I am honored to have been the recipient of the 'Science and Technology Innovation Award' for year 2025

01/06/2024
Science and Technology Innovation Award – Beijing University of Technology

I am honored to have been the recipient of the 'Science and Technology Innovation Award' for year 2024

04/09/2020
Chinese Government Scholarship – Beijing University of Technology

I am the recipient of a fully funded Chinese Government Scholarship for my PhD studies, covering all tuition and research expenses.

CONFERENCES & SEMINARS

12/07/2024 – 14/07/2024 Nanjing, China

9th International Conference on Signal and Image Processing (ICSIP 2024)

Successfully presented my paper at 2024 9th International Conference on Signal and Image Processing (ICSIP 2024)

18/03/2021 – 18/03/2021 Dera Ghazi Khan, Punjab, Pakistan

1st International Conference on Science Engineering & Technology (ICSET-21)

Certification of conference participation at MIR CHAKAR KHAN RIND UNIVERSITY OF TECHNOLOGY

13/11/2024 – 13/11/2024

Conference on Computer Science and Electrical / Electronic Engineering

Successfully presented the poster entitled Semantic segmentation of retinal anatomical structures and pathology in fundus Images at Beijing-Dublin International College (BDIC)

LANGUAGE SKILLS

Mother tongue(s): **URDU**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
CHINESE	A2	B1	A2	A2	A1
ENGLISH	C2	C2			C2
ARABIC	A2	A2	A2	A2	A1

Levels: A1 and A2: Basic user - B1 and B2: Independent user - C1 and C2: Proficient user

RECOMMENDATIONS

Professor. Dr. Feng Jinchao PhD Supervisor

Dean, Department of Information and Communication Engineering. Faculty of Information and Technology, Beijing University of Technology, P.R China.

Email fengjc@bjut.edu.cn

Tariq Mahmood Khan Professor

Professor, University of New South Wales

Email Tariq.khan@unsw.edu.au | **Phone** (+61) 420300045

Dr Muhammad Yaqub Mentor

Senior Researcher in Computer Science, Hunan University, Changsha, Hunan Province, P.R China, China (Work)

Email myaqub@hnu.edu.cn | **Phone** (+86) 17813086017

Imran Shabbir Professor

Department of Mathematics, UNIVERSITY OF KOTLI AZAD JAMMU AND KASHMIR, Pakistan

Email imran.shabbir@uokajk.edu.pk | **Phone** (+86) 18701617761